

Course:	COMP 2131	Written Assignment-1
Weight:	100 marks = 7% of final grade	Data Representation & Introduction to 'C'

Instructions

Complete the questions given below for Module 1, Module 2 and Module 3. Enter your answers directly into the document using **red text**. Once all the questions are completed submit it to the Open Learning Faculty Member.

Module-1 [30 marks]

Q1. What is the role of Operating System for the overall working of the Computer system? [10]

Q2. What is Cache Memory and why is it so important?[10]

Q3. How are processes and threads different? Explain with an example.[10]

Module-2 [45 marks]

Q1. Perform the following conversions:[10]

- Decimal number -91 to 8 bit two's complement representation
- Decimal number -398 into 16-bit two's-complement representation.

Q2. [15]

- Convert the string "0xC69A5" into ASCII.

b. What is the string corresponding to the following series of ASCII values:

0x54 0x68 0x69 0x73 0x20 0x69 0x73 0x20 0x41 0x53 0x43 0x49 0x49 0x21

c. Fill in the table below showing the effects of the different shift operations on single-byte quantities. Write each answer as two hexadecimal digits.[10]

x	x << 3	x >> 2	x >> 2
		(Logical)	(Arithmetic)
0xF0			
0x0F			
0xCC			
0x55			

Q3. Integer division on **most** machines is even slower than integer multiplication—requiring 30 or more clock cycles. Dividing by a power of two can also be performed using shift operations. Explain how this helps in performing division. Explain with an example. [10]

Q4. How is positive and negative infinity represented in the binary number?[10]

Module-3 [25 marks]

Q1. Explain the term masking. How is it useful for data representation? Explain the concept using an example. [5 marks]

Q2. [10 marks]

- a. For a floating-point format with a k -bit exponent and an n -bit fraction, give a formula for the smallest positive integer that cannot be represented exactly (because it would require an $n + 1$ -bit fraction to be exact).
- b. What is the numeric value of this integer for single-precision format ($k = 8$, $n = 23$)?

Q3. Write a 'C' code to check for even or odd number using bitwise operators. [10 marks]